

Rushil Chilakamarri

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PROFILE

- Computer Science graduate specializing in Data Science with hands-on experience building **full-stack, LLM-powered applications** – integrating cloud-hosted AI inference APIs into Python backends, designing **REST APIs**, and shipping interactive frontends. Strong foundation in **backend engineering, API design**, and applied **machine learning**, with a growing interest in agentic AI workflows and cloud-native deployment.

EDUCATION

Aurora's Technological and Research Institute <i>Bachelor of Technology in Computer Science - Data Science (CGPA: 7.67)</i>	Hyderabad 2022 – 2026
Sri Gayatri junior College <i>Mathematics, Physics, Chemistry</i>	Hyderabad 2020 – 2022

PROJECTS

MailSense – AI-Based Email Spam Detection and Tone Analyzer | *Flask, LLaMA, SQLite, Chart.js*

- Designed and delivered a **full-stack, LLM-powered application** end-to-end, integrating **cloud-hosted LLaMA inference APIs** into a **Python/Flask** backend for spam detection and tone classification.
- Built and documented **RESTful APIs** for real-time spam filtering and **tone analysis** of incoming emails.
- Integrated cloud-based AI inference APIs to improve scalability, efficiency, and response time, applying **prompt-based integration** patterns similar to foundation-model workflows.
- Integrated **SQLite** database for email storage, filtering history, and user management.
- Designed interactive dashboards using **Chart.js** to visualize email activity and classification insights.
- **Link:** github.com/Rushilch/Email

EcoVision – AI-Powered Environmental Risk Prediction System | *Streamlit, scikit-learn, GeoPandas, Pandas*

- Developed a **machine learning**-based platform to predict Urban Heat Islands (UHI), flood risks, and plastic waste hotspots.
- Implemented **Random Forest and Gradient Boosting** models for environmental risk classification.
- Integrated **Explainable AI** techniques such as SHAP and LIME to interpret predictions and feature importance.
- Built an interactive **Streamlit** dashboard for map-based visualization and environmental analytics.
- **Link:** github.com/Rushilch/EcoVision

VaaniVerse – Multilingual Voice to Sign Language System | *Python, OpenCV, MediaPipe, SpeechRecognition*

- Developed a **real-time** system to convert multilingual speech input into Indian Sign Language representations.
- Integrated **speech recognition** pipelines with preprocessing techniques for multilingual voice normalization.
- Built gesture rendering modules using **OpenCV and MediaPipe** achieving real-time sign visualization.
- Designed a desktop interface using **Tkinter** for real-time interaction and system control.
- **Link:** github.com/Rushilch/VaaniVerse

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++, C#, SQL
Full Stack & APIs: Flask, Django, FastAPI, REST API Design, React.js, Next.js, Angular, Streamlit, ASP.NET
AI/ML: LLM APIs, NLP, scikit-learn, PyTorch, TensorFlow, OpenCV, MediaPipe, LangChain & LangGraph
Cloud & DevOps: Docker, Git, Linux, Postman, AWS/EC2, Kubernetes
Libraries: Pandas, NumPy, Matplotlib
Databases: SQLite, MySQL, PostgreSQL, SQLyog, SQL sever
Core Skills: Machine Learning, Backend Development, API Design, Computer Vision

CERTIFICATIONS

CS50x: Introduction to Computer Science – Harvard University edx
Introduction to Deep Learning (6.S191) – MIT OpenCourseWare
100 Days of Code: The Complete Python Pro Bootcamp – Udemy
Python Mega Course: Learn Python in 60 Days, Build 20 Apps – Udemy
Machine Learning A-Z: AI, Python and R – Udemy
React - The Complete Guide (incl. Next.js, Redux) – Udemy
C# - Ultimate Guide: Beginner to Advanced Master Class (OOP) – Udemy
ASP.NET Core - True Ultimate Guide, Real Project – Udemy